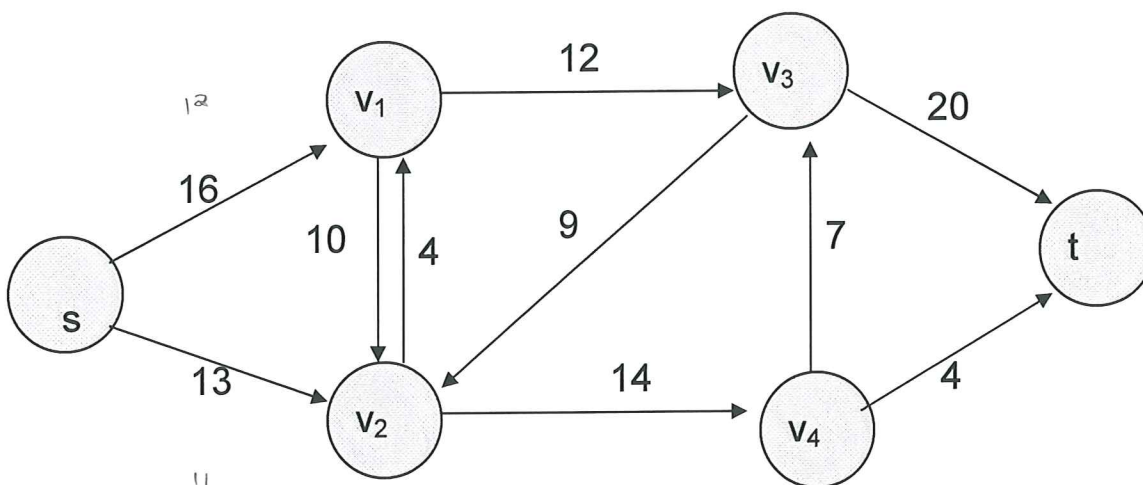


Name Brandon Buchanan

1) Find the Maximum Flow 23

2) Find the minimum cut which disconnects s from t. remove 12, 7, and 4



$$\max x_{3,t} + x_{4,t}$$

$$x_{s,1} \leq 16$$

$$x_{s,2} \leq 13$$

$$x_{1,2} \leq 10$$

$$x_{1,3} \leq 12$$

$$x_{2,1} \leq 4$$

$$x_{3,2} \leq 9$$

$$x_{3,t} \leq 20$$

$$x_{2,4} \leq 14$$

$$x_{4,3} \leq 7$$

$$x_{4,t} \leq 4$$

$$x_{s,1} = x_{1,2} + x_{1,3}$$

$$x_{s,2} = x_{3,1} + x_{2,4}$$

$$x_{1,2} = x_{2,1} + x_{2,4}$$

$$x_{2,1} = x_{1,2} + x_{1,3}$$

$$\cancel{x_{1,3} = x_{3,2} + x_{3,t}}$$

$$\checkmark x_{2,4} = x_{4,3} + x_{4,t}$$

$$x_{3,2} = x_{2,1} + x_{2,4}$$

$$\cancel{x_{4,3} = x_{3,t}}$$

$$x_{1,3} + x_{4,3} - x_{3,t} - x_{3,2} = 0$$

$$x_{s,1} + x_{2,1} - x_{1,2} - x_{1,3} = 0$$

$$x_{s,2} + x_{1,2} - x_{2,1} - x_{2,4} = 0$$